Chapter 7 - Superposition of Quasi-Parallel Plane Waves

- ↳ more complicated waves can be built → superposition
- ↳ phase velocity vs. group velocity
- ↳ Fourier theory
- ↳ dispersion

$$\vec{E}(\vec{r},t) = \sum_j \vec{E}_j e^{i(\vec{k}_j \cdot \vec{r} - \omega_j t)}$$

amplitude + phase

$$\vec{B}(\vec{r},t) = \sum_j \vec{B}_j e^{i(\vec{k}_j \cdot \vec{r} - \omega_j t)}$$

amplitude + phase

$$= \sum_j \frac{\vec{k}_j \times \vec{E}_j}{\omega_j} e^{i(\vec{k}_j \cdot \vec{r} - \omega_j t)}$$

$$\vec{S}(\vec{r},t) = \mathrm{Re}\{\vec{E}(\vec{r},t)\} \cdot \frac{\mathrm{Re}\{\vec{B}(\vec{r},t)\}}{\mu_0}$$

time average to exclude many short variations

limit to k's that point mostly in the same direction

$$\langle \vec{S} \rangle = I = \frac{n\epsilon_0 c}{2} \cdot \vec{E}(\vec{r},t) \cdot \vec{E}^*(\vec{r},t)$$

Group velocity vs. Phase Velocity

$$E_1 = \vec{E}_0 e^{i(\vec{k}_1 \cdot \vec{r} - \omega_1 t)} \qquad E_2 = \vec{E}_0 e^{i(\vec{k}_2 \cdot \vec{r} - \omega_2 t)}$$

$$v_{p1} = \frac{\omega_1}{k_1} \qquad v_{p2} = \frac{\omega_2}{k_2}$$

$$I(\vec{r},t) = \frac{n\epsilon_0 c}{2} \vec{E}_0 \cdot \vec{E}_0^* \left[e^{i(\vec{k}_1 \cdot \vec{r} - \omega_1 t)} + e^{i(\vec{k}_2 \cdot \vec{r} - \omega_2 t)} \right] \left[e^{-i(\vec{k}_1 \cdot \vec{r} - \omega_1 t)} + e^{-i(\vec{k}_2 \cdot \vec{r} - \omega_2 t)} \right]$$

$$I(\vec{r},t) = \frac{n\epsilon_0 c}{2} \vec{E}_0 \cdot \vec{E}_0^* \left(2 + e^{i(\vec{k}_1 \cdot \vec{r} - \omega_1 t - \vec{k}_2 \cdot \vec{r} + \omega_2 t)} + e^{i(\vec{k}_2 \cdot \vec{r} - \omega_2 t - \vec{k}_1 \cdot \vec{r} + \omega_1 t)} \right)$$

$$I(\vec{r},t) = \frac{n\epsilon_0 c}{2} \vec{E}_0 \cdot \vec{E}_0^* \left(2 + e^{i[(\vec{k}_2 - \vec{k}_1) \cdot \vec{r} - (\omega_2 - \omega_1) t]} + e^{-i[(\vec{k}_2 - \vec{k}_1) \cdot \vec{r} - (\omega_2 - \omega_1) t]} \right)$$

$$\cos(x) = \frac{e^{x} + e^{-x}}{2}$$

$$\sin(x) = \frac{e^{x} - e^{-x}}{2i}$$

$$I(\vec{r},t) = n\epsilon_0 c \cdot \vec{E}_0 \cdot \vec{E}_0^{*}\left(1 + \cos\left[\underbrace{(\vec{k}_2 - \vec{k}_1)}_{\Delta\vec{k}} \cdot \vec{r} - \underbrace{(\omega_2 - \omega_1)}_{\Delta\omega} t\right]\right)$$

$$v_p = \frac{\bar{\omega}}{\bar{k}}$$

$$\bar{\omega} = \frac{\omega_1 + \omega_2}{2}$$

average

$$\bar{k} = \frac{k_1 + k_2}{2}$$

$$v_g = \frac{\Delta\omega}{\Delta k} = \frac{d\omega}{dk}$$

for the jupyter example

$$v_p = \frac{\omega_1 + \omega_2}{k_1 + k_2} = \frac{1+2}{2+0.2} = \frac{3}{2.2}$$

$$v_p = 1.4$$

$$v_g = \frac{2-1}{2-0.2} = \frac{1}{1.8} = 0.6$$

$$v_g = 0.6$$

Example 7.1

Example 7.1

Determine the phase velocity and group velocity for the superposition of two plane waves in a plasma (see P2.7).

$$n_p(\omega) = \sqrt{1 - \frac{\omega_p^2}{\omega^2}} \qquad \omega > \omega_p \quad \text{so} \quad n_p(\omega) < 1$$

$$\frac{c}{n} = v = \frac{\omega}{k} \Rightarrow k = \frac{\omega \cdot n_p(\omega)}{c} \Rightarrow k_1 = \frac{\omega_1 \cdot n_p(\omega_1)}{c} \qquad k_2 = \frac{\omega_2 \cdot n_p(\omega_2)}{c}$$

$$v_p = \frac{\omega_2 + \omega_1}{k_2 + k_1} = \frac{\omega_2 + \omega_1}{\frac{\omega_2 n_p(\omega_2)}{c} + \frac{\omega_1 n_p(\omega_1)}{c}}$$

if ω_2 is nearly ω_1

$n_p(\omega_1) \approx n_p(\omega_2)$

$$v_p = \frac{c}{n_p(\omega)}$$

$$v_g = \frac{\Delta\omega}{\Delta k} \approx \frac{d\omega}{dk} = \left(\frac{dk}{d\omega}\right)^{-1} = \left[\frac{d}{d\omega}\left(\frac{\omega \cdot n_p(\omega)}{c}\right)\right]^{-1} = \left[\frac{d}{d\omega}\left(\frac{\omega}{c} \cdot \left(1 - \frac{\omega_p^2}{\omega^2}\right)^{1/2}\right)\right]^{-1}$$

physicist math!

$$= c\left[\underbrace{\frac{d}{d\omega}\left(\omega\left(1 - \frac{\omega_p^2}{\omega^2}\right)^{1/2}\right)}_{1}\right]^{-1}$$

$$\downarrow$$

$$\left(1-\frac{\omega_p^2}{\omega^2}\right)^{1/2} + \cancel{\omega}\cdot\left(+\frac{1}{\cancel{2}}\right)\left(1-\frac{\omega_p^2}{\omega^2}\right)^{-1/2}\cdot(\cancel{+2})\,\frac{\omega_p^2}{\omega^{\cancel{3}\,2}}$$

$$c\left[\left(1-\frac{\omega_p^2}{\omega^2}\right)^{1/2}\frac{\left(1-\frac{\omega_p^2}{\omega^2}\right)^{1/2}+\frac{\frac{\omega_p^2}{\omega^2}}{\left(1-\frac{\omega_p^2}{\omega^2}\right)^{1/2}}}{\left(1-\frac{\omega_p^2}{\omega^2}\right)^{1/2}}\right]^{-1}$$

← get common denom

$$= c\,\frac{\left(1-\frac{\omega_p^2}{\omega^2}\right)^{1/2}}{1-\cancel{\frac{\omega_p^2}{\omega^2}}+\cancel{\frac{\omega_p^2}{\omega^2}}}$$

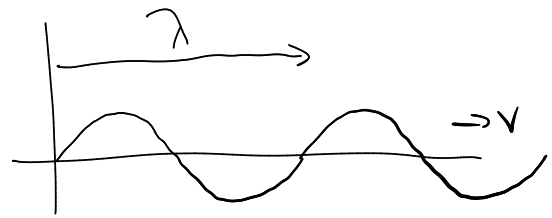
← spent my inverse power

$$V_g = c\cdot n_p(\omega)$$

Fourier Analysis

periodic waves → a wave that repeats itself over time

harmonic waves → sin/cosine wave



$$\cos\alpha + \cos\beta \neq \cos\gamma$$

$$= 2\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)$$

$$f(t) = \sum_{m=0}^{\infty} a_m \cdot \cos(m \cdot w \cdot t) + \sum_{m=0}^{\infty} b_m \sin(mwt)$$

} Fourier series

complete basis set

↳ Bessel Functions

↳ Hermite polynomials

↳ Legendre polynomials

↳ Laguerre polynomials

$$\boxed{f(t) = \frac{a_0}{2} + \sum_{m=1}^{\infty} a_m \cos(mwt) + \sum_{m=1}^{\infty} b_m \sin(mwt)}$$

$$\int_0^T f(t)\,dt = \underbrace{\int_0^T \frac{a_0}{2}\,dt}_{\frac{a_0 T}{2}} + \underbrace{\int_0^T \text{all the sines/cosines } dt}_{=0 \text{ if we are clever about } T}$$

We "know" what $f(t)$ is

we make $w = \frac{2\pi}{T} \rightarrow T = \frac{2\pi}{w}$

$$a_0 = \frac{2}{T}\int_0^T f(t)\,dt \quad \checkmark$$

Now for the infinitely many others:

$$\int_0^T f(t)\cos(n\omega t)\,dt = \underbrace{\int_0^T \frac{a_0}{2}\cos(n\omega t)\,dt}_{=0} + \underbrace{\sum_{m=1}^{\infty}\int_0^T a_m \cos(m\omega t)\cos(n\omega t)\,dt}_{\substack{=0 \quad m\neq n \\ =\frac{T}{2} \quad m=n}} + \underbrace{\sum_{m=1}^{\infty}\int_0^T b_m \sin(m\omega t)\cdot\cos(n\omega t)\,dt}_{=0}$$

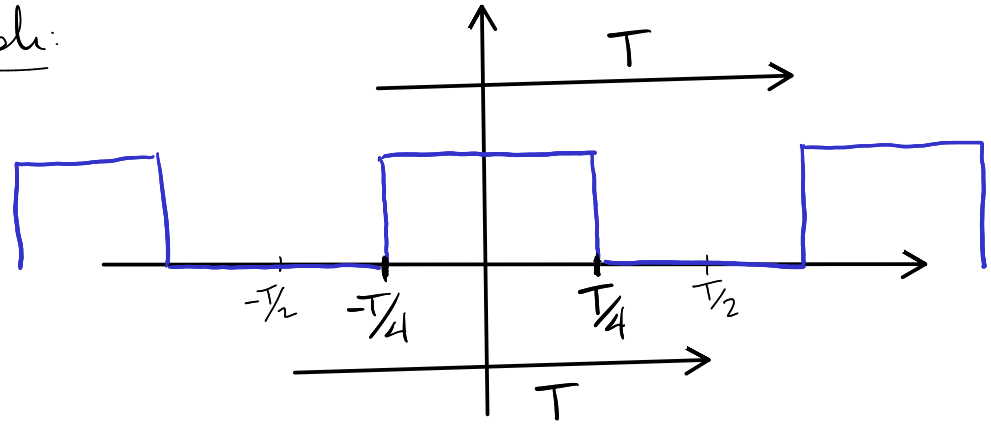
$$\int_0^T f(t)\cos(n\omega t)\,dt = \frac{T}{2}a_n$$

$$* \quad a_n = \frac{2}{T}\int_0^T f(t)\cdot\cos(n\omega t)\,dt$$

also I can do a similar thing to find b_n

$$* \quad b_n = \frac{2}{T}\int_0^T f(t)\cdot\sin(n\omega t)\,dt$$

Example:



$$f(t) = \begin{cases} 0 & -T/2 < t < -T/4 \\ 1 & -T/4 < t < +T/4 \\ 0 & T/4 < t < +T/2 \end{cases}$$

$$a_0 = \frac{2}{T}\int_{-T/2}^{+T/2} f(t)\,dt = \frac{2}{T}\underbrace{\int_{-T/4}^{+T/4} 1\,dt}_{t\Big|_{-T/4}^{+T/4}}$$

$$= \frac{2}{T}\left[\frac{T}{4} + \left(+\frac{T}{4}\right)\right]$$

$$= \frac{2}{T}\cdot\frac{T}{2}$$

$$= 1$$

$$a_n = \frac{2}{T}\int_{-T/2}^{T/2} f(t)\cdot\cos(n\omega t)\,dt$$

$$a_n = \frac{2}{T}\int_{-T/4}^{T/4} 1\cdot\cos(n\omega t)\,dt$$

$$= \frac{2}{T}\left(\frac{\sin(n\omega t)}{n\omega}\right)_{-T/4}^{+T/4}$$

$$= \frac{2}{Tn\omega}\left(\sin\left(n\omega\frac{T}{4}\right) - \sin\left(n\omega\left(-\frac{T}{4}\right)\right)\right)$$

$$\omega = \frac{2\pi}{T}$$

$$= \frac{\cancel{2}}{\cancel{T}\,n\left(\frac{2\pi}{\cancel{T}}\right)}\left(\sin\left(n\cdot\frac{\cancel{2}\pi}{\cancel{T}}\cdot\frac{\cancel{T}}{\cancel{4}_2}\right) - \sin\left(n\frac{\cancel{2}\pi}{\cancel{T}}\cdot\left(-\frac{\cancel{T}}{\cancel{4}_2}\right)\right)\right)$$

$\sin(-x) = -\sin(x)$

$\cos(-x) = \cos(x)$

$$\boxed{a_n = \frac{2}{n\pi}\cdot\sin\left(\frac{n\pi}{2}\right)} \xrightarrow[n \text{ for } m]{\text{just swap}} a_m = \frac{2}{m\pi}\cdot\sin\left(\frac{m\pi}{2}\right)$$

$$f(t) = \frac{a_0}{2} + \sum_{m=1}^{\infty} a_m \cos(m\omega t) + \sum_{m=1}^{\infty} b_m \sin(m\omega t)$$

$$b_n = \frac{2}{T}\int_{-T/2}^{T/2} f(t)\cdot \sin(n\omega t)\,dt$$

$$b_n = \frac{2}{T}\int_{-T/4}^{T/4} \sin(n\omega t)\,dt = 0$$

prove this!

So,

$$f(t) = \frac{1}{2} + \sum_{m=1}^{\infty} \frac{2}{m\pi}\cdot \sin\left(\frac{m\pi}{2}\right)\cdot \cos(m\omega t)$$

What about w/ complex exponentials?

$$f(t) = \sum_{m=0}^{\infty} a_m \cos(m\omega t) + \sum_{m=0}^{\infty} b_m \sin(m\omega t)$$

$$\cos(m\omega t) = \mathrm{Re}\left\{e^{-im\omega t}\right\} = \frac{e^{-im\omega t} + e^{+im\omega t}}{2}$$

fold in to a_m

$$\sin(m\omega t) = \mathrm{Im}\left\{e^{-im\omega t}\right\} = \frac{e^{-im\omega t} - e^{+im\omega t}}{2i}$$

fold in to b_m

$$f(t) = \sum_{m=0}^{\infty} a_m e^{-im\omega t} + \sum_{m=0}^{\infty} a_m e^{+im\omega t} + \sum_{m=0}^{\infty} b_m e^{-im\omega t} + \sum_{m=0}^{\infty} b_m e^{+im\omega t}$$

$$= \sum_{m=0}^{\infty} \underbrace{(a_m + b_m)}_{c_m} e^{-im\omega t} + \sum_{m=0}^{\infty} \underbrace{(a_m + b_m)}_{c_m} e^{+im\omega t}$$

$$m = -m'$$

$$= \sum_{m=0}^{\infty} c_m e^{-im\omega t} + \sum_{m'=0}^{-\infty} c_m e^{-im'\omega t}$$

$$m' = m$$

$$= \sum_{m=0}^{\infty} c_m e^{-im\omega t} + \sum_{m=0}^{-\infty} c_m e^{-im\omega t}$$

$$\boxed{f(t) = \sum_{m=-\infty}^{+\infty} c_m e^{-im\omega t}}$$

Now, we need to find c_m

$$\int_T f(t)\, e^{+in\omega t}\, dt = \sum_{m=-\infty}^{\infty} c_m \underbrace{\int_T e^{-im\omega t}\, e^{+in\omega t}\, dt}_{\begin{cases} = T & \text{if } m=n \\ = 0 & \text{if } m \neq n \end{cases}} = c_n \cdot T$$

$$T\delta_{nm}$$

$$c_n = \frac{1}{T}\int_T f(t)\, e^{+in\omega t}\, dt$$

$$\boxed{c_m = \frac{1}{T}\int_T f(t)\, e^{+im\omega t}\, dt}$$

$$\boxed{f(t) = \sum_{m=-\infty}^{+\infty} c_m e^{-im\omega t}}$$

Kronicker delta

$$\delta_{nm} = \begin{cases} 0 & n \neq m \\ 1 & n = m \end{cases}$$

$$\int_0^T \sin(n\omega t)\sin(m\omega t)\, dt = \frac{T}{2}\delta_{nm}$$

$$\int_0^T \cos(n\omega t)\cos(m\omega t)\, dt = \frac{T}{2}\delta_{nm}$$

$$\int_0^T \cos(n\omega t)\cdot\sin(m\omega t)\, dt = 0$$

PLOT THIS AND SEE

Now, what if we stop adding frequencies that differ by integer amounts and add <u>all</u> frequencies

$$f(t) = \underbrace{\sum_{-\infty}^{+\infty} c_n e^{-in\omega t} \cdot \underbrace{\Delta\omega}_{=1}}_{\text{Riemann Sum}} \longrightarrow f(t) = \int_{-\infty}^{\infty} g(\omega) e^{-i\omega t} \, d\omega$$

$\cancel{c_n}$

$g(\omega)$ is some <u>continuous</u> <u>function</u>

$$1 = \Delta\omega = n\omega - (n-1)\omega$$

$$\Delta\omega = \omega = \frac{2\pi}{T}$$

$\hookrightarrow \infty \leftarrow$ <u>not</u> <u>periodic</u>

$\downarrow$

$d\omega$

so, $f(t)$ is not periodic; it is now any function of t.

So our goal is to find the function $g(\omega)$ that satisfies our $f(t)$ and this integral.

we want to find $g(\omega)$

we know this one

$$f(t) = \int_{-\infty}^{\infty} g(\omega) e^{-i\omega t} \, d\omega$$

what function?

$$e^{+i\omega' t}$$

$$\int_{-\infty}^{\infty} f(t) \underline{\quad ? \quad} \, dt = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(\omega) e^{-i\omega t} \underline{\quad ? \quad} \, d\omega \, dt$$

$$\int_{-\infty}^{\infty} f(t) e^{+i\omega' t} \, dt = \int_{-\infty}^{\infty} g(\omega) \underbrace{\left[\int_{-\infty}^{\infty} e^{-i\omega t} \cdot e^{+i\omega' t} \, dt \right]}_{= 2\pi \cdot \delta(\omega - \omega')} d\omega$$

Dirac delta

$$\delta(\omega - \omega') = \begin{cases} 0 & \omega \neq \omega' \\ \infty & \omega = \omega' \end{cases}$$

$$= \int_{-\infty}^{\infty} g(\omega) \cdot 2\pi \cdot \delta(\omega - \omega') \, d\omega$$

$$\int_{-\infty}^{\infty} \delta(\omega - \omega') \, d\omega = 1$$

$$\int_{-\infty}^{\infty} f(t)\, e^{+i\omega' t}\, dt = 2\pi \underbrace{\int_{-\infty}^{\infty} g(\omega)\cdot\delta(\omega-\omega')\, d\omega}_{g(\omega')}$$

$$\int_{-\infty}^{\infty} f(t)\, e^{+i\omega' t}\, dt = 2\pi\, g(\omega')$$

day 37

260415 W

drop the primes

$$\boxed{g(\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty} f(t)\, e^{+i\omega t}\, dt}$$

Fourier Transform

$$\boxed{f(t) = \int_{-\infty}^{\infty} g(\omega)\, e^{-i\omega t}\, d\omega}$$

Inverse Fourier Transform

$$f(t) = \mathrm{IFT}\left[g(\omega)\right]$$

$$\mathrm{FT}\left[\mathrm{IFT}\left[g(\omega)\right]\right] = g(\omega)$$

$$f(t) = \mathrm{IFT}\left[\mathrm{FT}\left[f(t)\right]\right]$$

Spacial Fourier Transform

our wavefunction $\rightarrow f(kx - \omega t)$

$\omega = \frac{2\pi}{T}$ $\quad \omega \rightarrow k \quad$ $k = \frac{2\pi}{\lambda}$ $\Rightarrow$

$t \rightarrow x$

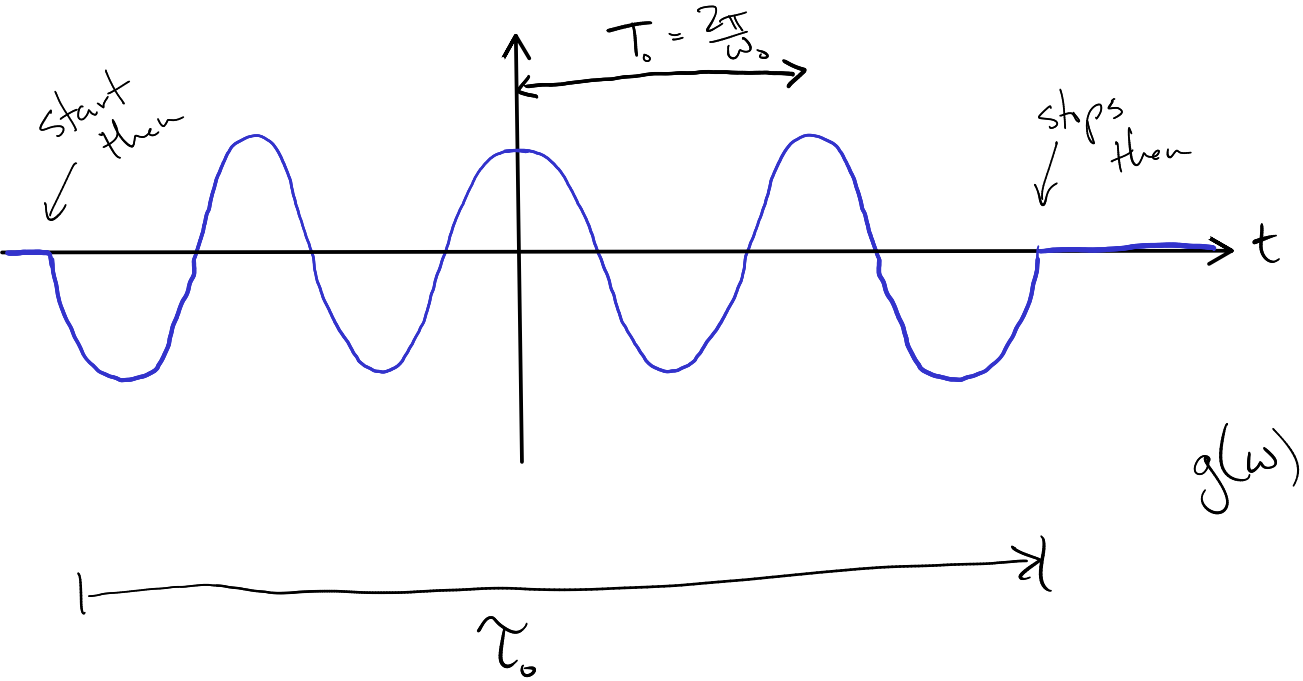
angular frequency

spacial frequency

$$f(x) = \int_{-\infty}^{\infty} g(k) e^{-ikx} dk \leftarrow \text{Inverse Fourier}$$

$$g(k) = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(x) e^{+ikx} dx \leftarrow \text{Fourier}$$

Ex. Harmonic-ish but finite.



$$f(t) = \begin{cases} e^{-i\omega_0 t} & -\frac{\tau_0}{2} < t < \frac{\tau_0}{2} \\ 0 & \text{elsewhere} \end{cases}$$

$$g(\omega) = \frac{1}{2\pi} \int_{-\frac{\tau_0}{2}}^{+\tau_0/2} \underbrace{e^{-i\omega_0 t} \cdot e^{+i\omega t}} \, dt$$

$$= \frac{1}{2\pi} \int_{-\tau/2}^{\tau/2} e^{it(\omega-\omega_0)} \, dt$$

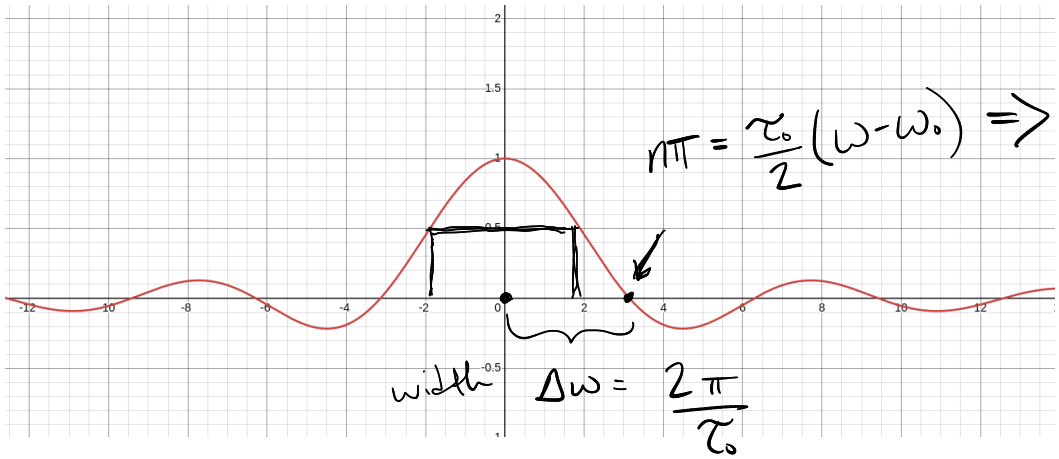
$$= \frac{1}{2\pi} \frac{1}{i(\omega-\omega_0)} e^{it(\omega-\omega_0)} \Big|_{-\frac{\tau_0}{2}}^{+\frac{\tau_0}{2}}$$

$$g(\omega) = \frac{1}{\pi} \frac{1}{(\omega-\omega_0)} \left[\frac{e^{i\frac{\tau_0}{2}(\omega-\omega_0)} - e^{-i\frac{\tau_0}{2}(\omega-\omega_0)}}{2i} \right]$$

$$g(\omega) = \frac{\sin\left(\frac{\tau_0}{2}(\omega-\omega_0)\right)}{\pi(\omega-\omega_0)} \frac{\left(\frac{\tau_0}{2}\right)}{\left(\frac{\tau_0}{2}\right)}$$

$$g(\omega) = \frac{\tau_0}{2\pi} \cdot \underbrace{\frac{\sin\left(\frac{\tau_0}{2}(\omega-\omega_0)\right)}{\frac{\tau_0}{2}(\omega-\omega_0)}}_{\operatorname{sinc}\left(\frac{\tau_0}{2}(\omega-\omega_0)\right)}$$

$$\operatorname{sinc}(x) = \frac{\sin(x)}{x}$$



$$n\pi = \frac{\tau_0}{2}(\omega-\omega_0) \Rightarrow \omega = \omega_0 + \frac{2n\pi}{\tau_0}$$

width $\Delta\omega = \dfrac{2\pi}{\tau_0}$

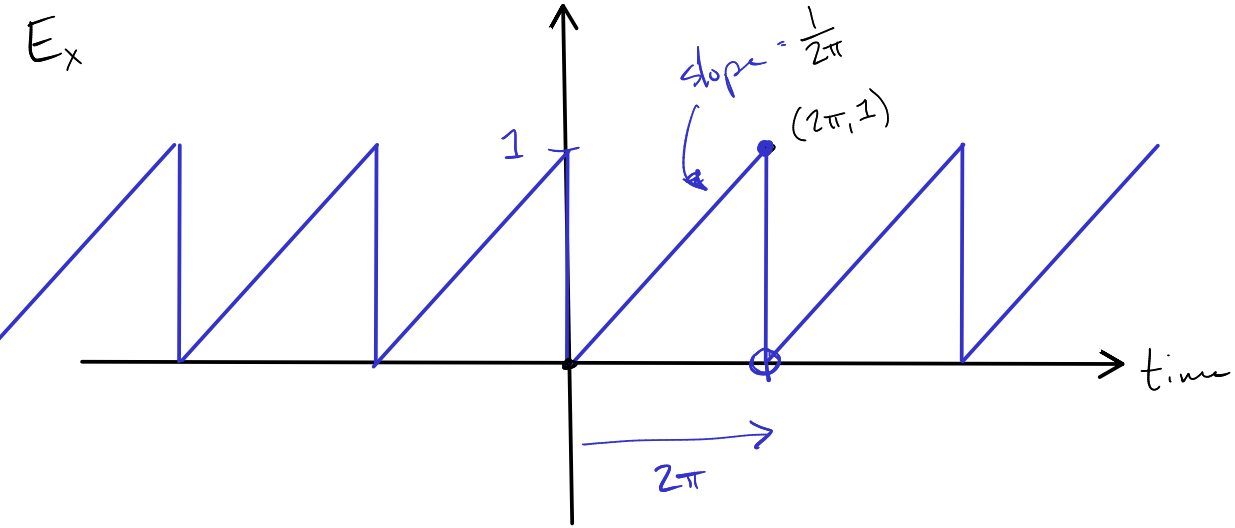
$$\sin(0) \equiv 1$$

$$\Delta\omega = \frac{2\pi}{\tau_0} \longleftrightarrow \omega = \frac{2\pi}{T} = 2\pi\nu$$

$$\Delta\omega = 2\pi\,\Delta\nu$$

where $\underbrace{\Delta\nu}_{\text{frequency bandwidth}} = \dfrac{1}{\tau_0}$

Ex



$$\boxed{c_m = \frac{1}{T}\int_T f(t)\, e^{+im\omega t}\, dt}$$

$$\boxed{f(t) = \sum_{m=-\infty}^{+\infty} c_m e^{-im\omega t}}$$

$f(t) = \frac{1}{2\pi} t$ over 2π

$$c_m = \frac{1}{2\pi}\int_0^{2\pi} \frac{t}{2\pi}\, e^{+im\omega t}\, dt$$

$$c_m = \frac{1}{(2\pi)^2}\int_0^{2\pi} t \cdot e^{im\omega t}\, dt$$

$$\int u\, dv = uv - \int v\, du$$

$u = t$

$du = dt$

$dv = e^{im\omega t}$

$v = \frac{1}{im\omega} e^{im\omega t}$

$$= \frac{1}{(2\pi)^2}\left(\left[\frac{t e^{im\omega t}}{im\omega}\right]_0^{2\pi} - \frac{1}{im\omega}\int_0^{2\pi} e^{im\omega t}\, dt \right)$$

$$= \frac{1}{4\pi^2}\left(\left[\frac{t e^{im\omega t}}{im\omega}\right]_0^{2\pi} + \frac{1}{m^2\omega^2}\left[e^{im\omega 2\pi} - 1\right] \right)$$

$$= \frac{im\omega}{im\omega}\left[\frac{2\pi e^{im\omega 2\pi}}{4\pi^2 im\omega}\right] + \frac{e^{im\omega 2\pi} - 1}{4\pi^2 m^2 \omega^2}$$

$$= \frac{-2\pi im\omega e^{im\omega 2\pi} + e^{im\omega 2\pi} - 1}{4\pi^2 m^2 \omega^2}$$

$$= \frac{-1 + e^{2\pi im\omega}(1 - 2\pi im\omega)}{4\pi^2 m^2 \omega^2}$$